

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & -2 & 0 \\ -1 & 3-\lambda & -1 \\ 2 & 3 & 1-\lambda \end{vmatrix}$$

$$a_{11}M_{11} - a_{12}M_{12} - a_{13}M_{13}$$

$$(-1-\lambda) \begin{vmatrix} 3-\lambda & -1 \\ 3 & 1-\lambda \end{vmatrix}$$

$$(3-\lambda)(1-\lambda) - (-1) \cdot 3 = (3-\lambda)(1-\lambda) + 3 = \lambda^2 - 4\lambda + 6$$

první člen: $(-1-\lambda)(\lambda^2 - 4\lambda + 6)$

$$-(-2) \begin{vmatrix} -1 & -1 \\ 2 & 1-\lambda \end{vmatrix}$$

$$(-1) \cdot (1-\lambda) - (-1) \cdot 2 = -1 + \lambda + 2 = \lambda + 1$$

druhý člen $2(\lambda + 1)$

třetí člen 0

rovnice: $(-1-\lambda)(\lambda^2 - 4\lambda + 6) + 2(\lambda + 1) = -(\lambda - 2)^2(\lambda + 1)$

vlastní čísla: $\lambda_1 = 2$, alg. nás. 2
 $\lambda_2 = -1$, alg. nás. 1

Pro $\lambda = 2$

$$(A - 2I)x = 0: \begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix}$$

$$E_2 = \text{span}\{(2, -3, -5)\}$$

$$\dim = 1$$

Pro $\lambda = -1$

$$\begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 2 & 3 & 2 \end{pmatrix}$$

$$E_{-1} = \text{span}\{(2, 0, -1)\}$$

$$(A - 2I)v_2 = v_1 \quad \text{hde} \quad v_1 = (2, -3, -5)$$

$$v_2 = (1, 0, 1)$$

$$U_1 = \text{span}\{v_1, v_2\} = \text{span}\{(2, -3, -5), (1, 0, 1)\}$$

$$B_{U_1} = (v_1, v_2): \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$$U_2 = \text{span}\{v_1, w\}, \quad w = (2, 0, -1)$$

$$\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$